

第九章作业

9.5

$$\text{由 } \cos \theta = \sin(\theta + 90^\circ),$$

$$\text{则 } V_2 = 50 \cos(\omega t - 30^\circ) = 50 \sin(\omega t - 30^\circ + 90^\circ) = 50 \sin(\omega t + 60^\circ)$$

$$\text{又由 } V_1 = 45 \sin(\omega t + 30^\circ)$$

则相差为 30° , V_1 是滞后的

9.51

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j \times 2 \times 0.1} = -j5, \quad Z_L = j\omega L = j \times 2 \times 0.5 = j$$

$$Z = -j5 + j + 2 = 2 - j4, \quad I = \frac{90}{Z} \angle 0^\circ = 45 \angle 0^\circ,$$

$$\text{由 } I = \frac{1}{1 - j5 + j + 2} I_s, \text{ 则 } I_s = I(3 - j4), \text{ 代入 } I = 45 \angle 0^\circ, \text{ 可得,}$$

$$I_s = 45(3 - j4), \quad |I_s| = 45\sqrt{3^2 + 4^2} = 225, \quad \angle I_s = -\tan^{-1} \frac{4}{3} = -53.13^\circ,$$

$$\text{则 } I_s = 225 \angle -53.13^\circ,$$

$$i(t) = 225 \cos(\omega t - 53.13^\circ) \text{ A}$$

9.54

左边阻抗: $2Z$, 右边阻抗: Z , 右边部分等效阻抗为 $Z = 1 - j$

$$\text{右侧电压 } V_1 = 30Z = 30(1 - j), \quad V_2 = 2V_1 = 60(1 - j)$$

$$\text{由 KVL: } V_2 + V_3 + V_1 = 0, \text{ 则 } V_3 = -V_1 - V_2 = -30(1 - j) - 60(1 - j) = -90(1 - j) = -90 + 90j$$

$$|V_3| = \sqrt{(-90)^2 + 90^2} = 90\sqrt{2} = 127.28, \quad \theta = 135^\circ$$

$$\text{则 } V_3 = 127.28 \angle 135^\circ \text{ V}$$

9.66

左支路: $Z_1 = 20 - j5$, 右支路 $Z_2 = 40 + j10$,

$$Z_T = (20 - j5) \parallel (40 + j10) = \frac{(20 - j5)(40 + j10)}{60 + j5}, \text{ 则 } Z_T = 14.069 - j1.172 = 14.118 \angle -4.76^\circ \Omega$$

$$I = \frac{V}{Z_T} = \frac{115 \angle 90^\circ}{14.118 \angle -4.76^\circ} = 8.146 \angle 94.76^\circ \text{ A}$$

$$\text{左支路 } I_1 = \frac{40 + j10}{60 + j5} I = \frac{8 + j2}{12 + j} I, \text{ 右支路 } I_2 = \frac{20 - j5}{60 + j5} = \frac{4 - j}{12 + j} I$$

则

$$V_{ab} = -20I_1 + j10I_2 = -20\left(\frac{8 + j2}{12 + j}\right)I + j10\left(\frac{4 - j}{12 + j}\right)I$$

$$= \frac{-(160 + j40)}{12 + j} I + \frac{10 + j40}{12 + j} I$$

$$= \frac{-150}{12 + j} I$$

$$\text{则 } V_{ab} = (12.457 \angle 175.24^\circ)(8.146 \angle 94.76^\circ) = 101.47 \angle 270^\circ \text{ V}$$

9.67

$$(a) \quad Z_L = j\omega L = j \times 10^3 \times 20 \times 10^{-3} = j20$$

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j \times 10^3 \times 12.5 \times 10^{-6}} = -j80$$

$$Z_{in} = 60 + j20 \parallel 60 - j80 = 60 + \frac{j20(60 - j80)}{j20 + 60 - j80} = 63.33 + j23.33$$

化为极坐标为 $Z_{in} = 67.494 \angle 20.22^\circ \Omega$

$$R1) \quad Y_{in} = \frac{1}{Z_{in}} = \frac{1}{67.494 \angle 20.22^\circ} = 14.8 \angle -20.22^\circ \text{ mS}$$

$$(b) \quad Z_L = j\omega L = j \times 10^3 \times 10 \times 10^{-3} = j10$$

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j \times 10^3 \times 20 \times 10^{-6}} = -j50, \quad 30 \parallel 60 = 20,$$

$$Z_{in} = -j50 + 20 \parallel (40 + j10) = -j50 + \frac{20(40 + j10)}{60 + j10} = 13.5 - j48.92$$

$$Z_{in} = 50.75 \angle -74.56^\circ \Omega$$

$$Y_{in} = \frac{1}{Z_{in}} = \frac{1}{50.75 \angle -74.56^\circ} = 19.704 \angle 74.56^\circ \text{ mS}$$

$$\text{即 } Y_{in} = 5.246 + j18.993 \text{ mS}$$